

Question

$[\text{Cu}(\text{CH}_3\text{CN})_4](\text{PF}_6)$, $\text{Na}(\text{S}_2\text{CNR}_2)$ and $[(^n\text{C}_4\text{H}_9)_4\text{N}](\text{BH}_4)$ were mixed in acetonitrile according to stoichiometric ratio, stirred at room temperature for 1 hour under nitrogen protection, the solvent was evaporated, and the residue was recrystallized from chloroform to obtain a yellow solid. The yellow solid is a Cu(I) polynuclear cluster with the chemical formula $[\text{Cu}_8\text{H}(\text{S}_2\text{CNR}_2)_x](\text{PF}_6)_y$, and the hydride ion is encapsulated in the cluster; elemental analysis shows that the mass fractions of each element in the cluster are: C, 23.24%; H, 4.14%; N, 5.64%

The following statement that is correct is:

1. R is methyl
2. In the cation, there are two kinds of environments for Cu (ratio is 1:1), one of which Cu is connected to 6 Cu, and the other is connected to 3 Cu. Then Cu constitutes a tetracapped tetrahedron.
3. In the cation, each Cu is coordinated by 3 S, then each S is coordinated by 2 Cu.
4. The cluster can also be obtained by mixing $[\text{Cu}(\text{CH}_3\text{CN})_4](\text{PF}_6)$, $\text{Cu}(\text{S}_2\text{CNR}_2)_2$ and $[(^n\text{C}_4\text{H}_9)_4\text{N}](\text{BH}_4)$ in acetonitrile according to stoichiometric ratio. The sum of the stoichiometric numbers of the reaction equation (the coefficient is balanced to the simplest integer ratio, and the species in the equation need to reflect the real structural form) is 41

A. All other options are incorrect

B. 2, 3

C. 2, 4

D. 1, 2, 4

E. 1, 2

F. 2, 3

G. 1, 2, 3, 4

H. 2

I. 3, 4

J. 1, 3, 4

K. 2, 3, 4

L. 1, 2, 3

M. 1

N. 4

Answer

Correct Answer: **K**

Detailed Explanation

The number of carbon atoms in the ligand is: $14.01/5.64\% \times 23.24\%/12.01 = 4.81$, R contains 2 carbons

The number of hydrogen atoms in the ligand is much larger than that outside the ligand, approximately: $14.01/5.64\% \times 4.14\%/1.008 = 10.2$, that is, R contains 5 hydrogens

So $R = Et$, statement 1 is incorrect

CHECKPOINT

2 PTS

R is ethyl

According to charge balance, we have: $x + y = 7$

According to the mass fraction, we have: $[63.55 \times 8 + 1.008 + 148.3x + 145.0 \times y] \times 5.64$

Solving for: $x = 6, y = 1$

CHECKPOINT

2 PTS

$x = 6, y = 1$

Each Cu in the Cu tetrahedron is connected to 3 Cu, so we can first assemble 4 Cu tetrahedra. To make the 4 Cu in the cation connected to 6 Cu, we can combine them into another tetrahedron, which is a capped tetrahedral type, statement 2 is correct

CHECKPOINT

2 PTS

Cu in the cation forms a capped tetrahedron

The cation contains a total of $8Cu + 12S$, $8 \times 3/12 = 2$, so each S is coordinated by 2 Cu, statement 3 is correct

CHECKPOINT

1 PTS

Each S is coordinated by 2 Cu

Let the coefficient of $\text{Cu}(\text{S}_2\text{CNEt}_2)_2$ be 3a, then the coefficient of $[\text{Cu}(\text{CH}_3\text{CN})_4](\text{PF}_6)$ is 5a, $[(^n\text{C}_4\text{H}_9)_4\text{N}]^+$ needs to form a salt with $(\text{PF}_6)^-$, so the coefficient of $[(^n\text{C}_4\text{H}_9)_4\text{N}](\text{BH}_4)$ is 4a. Cu is reduced and H_2 is released, and the remaining boron element should exist in the form of B_2H_6 .

CHECKPOINT

1 PTS

The remaining boron element should exist in the form of B_2H_6



CHECKPOINT

2 PTS



The sum of the coefficients is 41, statement 4 is correct

Select K